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Herman Wandabwa, PhD

Senior AI Scientist | GenAI & Agentic Systems

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Senior AI Scientist with a PhD specialising in generative AI and agentic systems, with 7+ years translating research into production. Deep experience in LLM fine-tuning (Qwen, LLaMA), retrieval-augmented generation, and ReAct-style agent orchestration, backed by a strong applied research track record including a paper currently under review at ICML 2026. Rigorous experimental design and statistical validation paired with end-to-end MLOps delivery across financial services, insurance, and logistics. Effective communicator with senior stakeholders and a committed mentor of junior data scientists.

Skills

Generative AI & Agentic Systems:	LLM fine-tuning (Qwen-7B, LLaMA), prompt engineering, retrieval-augmented generation (RAG), ReAct agent reasoning loops, tool-use interfaces, orchestration logic, memory systems, context management, agentic evaluation frameworks, AI risk assessment and responsible AI governance
LLM Tooling & Frameworks:	Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, PEFT/LoRA, vLLM, FAISS
Machine Learning & Data Science:	Predictive modelling, propensity scoring, uplift modelling, customer segmentation, churn analytics, time-series forecasting, NLP (text classification, clustering, sentiment analysis, embeddings), self-supervised representation learning, mathematical optimisation
MLOps & Engineering:	End-to-end ML lifecycle, scalable deployment architectures, GCP, AWS, Azure ML, Spark, SQL (Snowflake), data pipeline engineering, feature engineering on large-scale data
Leadership & Stakeholder Engagement:	Mentoring and growing junior data scientists, code and model reviews, data storytelling, translating insights into business strategy, cross-functional collaboration in large engineering teams

Technical Experience

Senior AI Scientist — **January 2024 – Present**
Commonwealth Bank — *Melbourne, Australia*

Agentic AI & LLM Projects

- Designed a specialised Architect Agent (built in LangGraph) that ingests raw BPMN process definitions and automatically translates them into functional ReAct agentic applications, achieving ~40% lower input/output token usage than Roo Code, Cline, AutoGen, MetaGPT, and AFLOW variants evaluated under the same conditions. *Paper under review at ICML 2026.*
- Built an end-to-end Agentic AI Simulator, Evaluator and Mitigator framework for use by developers during the pre-deployment testing phase, supporting failure-mode diagnosis, reliability assessment, and targeted performance improvement for agents serving vulnerable customer populations; informs internal responsible AI governance for agent deployments.
- Fine-tuned and evaluated a Qwen-7B language model for financial-advice use cases, running rigorous experiments on response quality and domain accuracy and producing a reusable evaluation harness for downstream LLM projects.

Customer Analytics & Production ML

- Led customer targeting and segmentation using propensity scoring and behavioural modelling, delivering an 11% uplift in home-loan retention across a 500K+ customer base.
- Developed NLP-powered customer clustering for investor rental leads, achieving a 9% conversion increase through statistically validated segmentation and rigorous experimental design.
- Engineered scalable Spark pipelines transforming transaction-level data into customer behaviour features for production targeting models, reducing feature build time by ~50%.
- Mentored 4+ junior analysts in experimentation design, uncertainty quantification, and commercial framing of insights, contributing to lifted analytical standards across the team.

Data Scientist — **July 2022 – December 2023**
Commonwealth Bank — *Sydney, Australia*

- Pioneered self-supervised embedding techniques for customer transaction profiling, producing reusable features now deployed in 3+ production targeting systems.
- Developed customer uplift models and churn analytics for migrant segments, reducing attrition by 12% via validated targeting and intervention strategies.
- Designed a multi-objective optimisation framework for fraud-detection queue management using beam search, projecting $10M annual cost savings through improved resource allocation.
- Conducted rapid experimentation on Buy Now Pay Later (BNPL) adoption across demographic segments, with findings directly shaping product strategy.

Senior Analytics Specialist — **November 2021 – June 2022**
Tower Insurance — *Auckland, New Zealand*

- Built customer churn and uplift models using behavioural segmentation, lifting retention by 15% and operationalising targeting with marketing teams.
- Created an NLP-powered automation tool that reduced internal support query volume by 40%, demonstrating end-to-end prototype-to-production AI deployment.

Data Science Solutions Specialist **November 2020 – November 2021**
KiwiRail *Auckland, New Zealand*

- Designed and deployed a linear-programming model (PuLP, Google OR-Tools) for container allocation, boosting operational efficiency by 60%.
- Built time-series forecasting models for customer booking patterns, improving resource planning and reducing last-minute service disruptions by ~25%.
- Developed the company's first predictive model for on-time train performance, improving scheduling accuracy and lifting DIFOT (Delivery In Full, On Time) metrics.
- Applied NLP to incident reports to surface recurring failure modes, informing operational policy decisions and safety improvements.

Data Science Content Developer & Software Tutor (Casual) **February 2020 – August 2021**
Torrens University & Vision College *Adelaide, Australia / Auckland, NZ*

- Authored academic content covering Python, NLP, deep learning, computer vision, and model deployment for university-level applied data science instruction.

Data Scientist **June 2018 – December 2018**
Dexibit *Auckland, New Zealand*

- Built time-series forecasting models for museum visitor prediction and applied NLP sentiment analysis, integrating both into Dexibit's production SaaS platform.

Applied Research & Publications

In progress

- **Wandabwa, H.** (2026). *BPMN-to-ReAct: Token-Efficient Translation of Business Process Definitions into Agentic Applications.* Submitted to the International Conference on Machine Learning (ICML 2026). *[Under review]*
- **Wandabwa, H.** (2026). *An Agentic AI Simulator, Evaluator and Mitigator for Vulnerable Customers.* Applied research, Commonwealth Bank.

Selected peer-reviewed publications

- **Wandabwa, H., Naeem, M. A., Mirza, F., & Pears, R.** (2021). Topical affinity in short text microblogs. *Information Systems*, 96, 101662.
- **Wandabwa, H., Naeem, M. A., Mirza, F., & Pears, R.** (2021). Multi-interest semantic changes over time in short-text microblogs. *Knowledge-Based Systems*, 228, 107249.
- Nguyen, A., **Wandabwa, H.,** Rasco, A., & Le, L. A. (2021). A Framework for Designing Learning Analytics Information Systems. *Hawaii International Conference on System Sciences (HICSS)*, 1–10.
- **Wandabwa, H., Naeem, M. A., Mirza, F., Pears, R., & Nguyen, A.** (2020). Multi-interest User Profiling in Short Text Microblogs. *DESRIST 2020*, 154–168.

Selected Personal Projects and Research

Technical write-ups with accompanying code, published on Medium.

- **Agentic AI, LLMs & RAG Systems** — Multi-agent systems, RAG pipelines, and LLM fine-tuning across domains. Selected articles: What Agentic AI Actually Is – SIM Replacement (Part 1) ● Building a SIM-Swap Agent in LangGraph (Part 2) ● Fine-tuning LLaMA-2 on Safaricom Product FAQs ● RAG with MMR for Safaricom's Smart Assistant ● RAG Chatbot for Fitness.
- **Predictive Modelling for User Retention** — Customer churn prediction and uplift modelling for targeted intervention optimisation: Predicting Customer Churn Before It Happens ● Uplift Modelling to Target Churn Interventions.
- **Forecasting & Optimisation** — Demand forecasting, mathematical optimisation, and behavioural analytics: Predicting Daily Museum Visitors (Part 1) ● Museum Visitors (Part 2) ● Freight-Train Capacity Optimisation (Part 1) ● Freight Capacity with Real Data (Part 2) ● Wi-Fi Analytics for Premium Advertising.

Education

PhD in Computer and Information Sciences, *Auckland University of Technology, New Zealand* March 2017 — March 2021